(We discuss compound queries at a high level here, but we discuss them in detail in chapter 11.)

The compound queries are as follows:

- Boolean (`bool`)
- Constant score (`constant_score`)
- Function score (`function_score`)
- Boosting (`boosting`)
- Disjunction max (`dis_max`)

Of these, the `bool` query is the most commonly used, so we look at the `bool` query in action here.

2.5.1 Boolean (bool) query

A Boolean (`bool`) query is used to create sophisticated query logic by combining other queries based on Boolean conditions. A `bool` query expects the search to be built using a set of four clauses: `must`, `must_not`, `should`, and `filter`. The following listing shows the format of a `bool` query.

Listing 2.18 Format of a `bool` query with the expected clauses

```
GET books/_search
{
  "query": {
    "bool": {
      "must": [{ }],
      "must_not": [{ }],
      "should": [{ }],
      "filter": [{ }]
    }
  }
}
```

A bool query is a combination of conditional boolean clauses.

The criteria must match the documents.

The criteria must not match (no score contribution).

The query should match.

The query must match (no score contribution).

As you can see, the `bool` query expects one or more of the clauses `must`, `must_not`, `should`, and `filter` to define the criteria (see table 2.2). More than one criterion can be expressed with a combination of these clauses.

Table 2.2 Boolean (`bool`) query clauses

Clause	Explanation
`must`	The `must` clause means the search criteria in a query *must* match the documents. A positive match bumps up the relevancy score. We build a `must` clause with as many leaf queries as possible.
`must_not`	In a `must_not` clause, the criteria *must not* match the documents. This clause does not contribute to the score (it is run in a filter execution context; more on contexts in chapter 11).